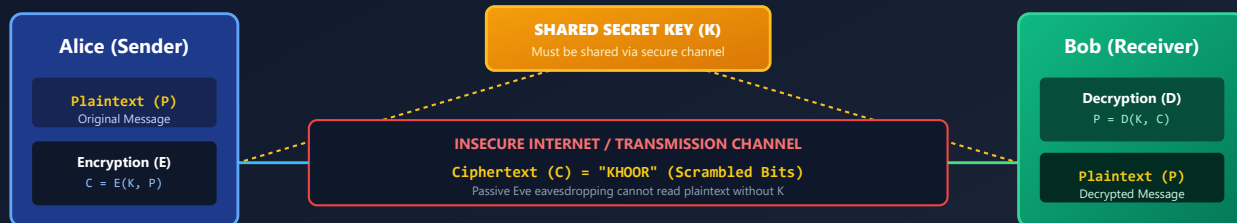


Module 10: Cryptography Foundations and Symmetric Ciphers

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CRYPTOGRAPHY FOUNDATIONS & SYMMETRIC CIPHERS

Confidentiality, Substitution vs Transposition Ciphers, DES & AES Block Cipher Architecture



CLASSICAL vs MODERN SYMMETRIC CRYPTOGRAPHY (SLIDES 73 - 76)

1. Substitution Ciphers

- Characters replaced by other symbols.

Monoalphabetic: Fixed 1-to-1 mapping. Caesar (Shift 3): HELLO -> KHOOR • **Polyalphabetic:** Character mapping changes based on position (Vigenère cipher). • Vulnerable to letter frequency analysis! 2. Transposition Ciphers • Characters retain identity; positions shuffle. • **Rail Fence Cipher:** Plaintext written in zig-zag diagonal rows and read row-by-row. • **Columnar Transposition:** Written in matrix grid and read out based on key order. • Frequency analysis shows standard English letter ratios. 3. Modern Block Ciphers • **DES (Data Encryption Standard):** 64-bit block, 56-bit key, 16 Feistel rounds. (Broken via brute force due to small key). • **AES (Advanced Encryption Standard):** 128-bit block, 128/192/256-bit key. SPN rounds (SubBytes, ShiftRows, MixColumns). • Global standard for military & TLS data.

1. Cryptographic Principles & Security Dimensions

Cryptography ensures secure communication across four pillars: Confidentiality, Integrity, Authentication, and Non-repudiation.

Symmetric Cryptography Architecture

A single secret key K performs both encryption and decryption: $C = E(K, P)$ and $P = D(K, C)$. Requires pre-shared key establishment.

2. Classical Ciphers: Substitution vs Transposition

Cipher Family	Structural Mechanism	Vulnerability Profile
Substitution (Monoalphabetic)	Replaces plaintext letters with static fixed alphabet mapping (e.g., Caesar shift)	Vulnerable to natural letter frequency analysis (e.g., English 'E', 'T')
Transposition (Permutation)	Shuffles letter positions without modifying underlying alphabet characters (Rail Fence, Columnar)	Preserves original character frequency histograms

3. Modern Block Ciphers: DES vs AES

Parameter	DES	AES
Block Size	64 bits	128 bits
Effective Key Length	56 bits (Insecure against modern GPUs)	128, 192, 256 bits (Cryptographically unbreakable)
Round Design	16-Round Feistel Network	10 / 12 / 14 Rounds SPN (SubBytes, ShiftRows, MixColumns)